TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	72.1 ± 2.59
Correlation Stability	69.4 ± 4.57
Distribution Consistency	82.6 ± 3.95
Entropy Preservation	84.3 ± 1.62
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	80.0 ± 0.00
Privacy Risk	6.0 ± 2.79
Data Density	99.7 ± 0.26
Coverage Score	41.6 ± 9.13
Feature Accuracy	73.7 ± 3.26
Novelty Score	0.8 ± 0.02

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.240	-2.506	5.6%	OK
TB	3.299	1.978	-1.320	40.0%	Drift
DB	1.486	0.820	-0.666	44.8%	Drift
Alkphos	290.576	230.354	-60.222	20.7%	Warn
Sgpt	80.714	48.489	-32.224	39.9%	Drift
Sgot	109.911	55.758	-54.153	49.3%	Drift
TP	6.483	6.740	+0.257	4.0%	OK
ALB	3.142	3.364	+0.222	7.1%	OK
A/G Ratio	0.947	1.001	+0.054	5.7%	OK
Selector	1.286	1.373	+0.087	6.7%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	72.1 ± 2.59
Correlation Stability (*2)	69.4 ± 4.57
Distribution Consistency (*3)	82.6 ± 3.95
Entropy Preservation (*4)	84.3 ± 1.62
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	80.0 ± 0.00
Privacy Risk (*7)	6.0 ± 2.79
Data Density (*8)	99.7 ± 0.26
Coverage Score (*9)	41.6 ± 9.13

Feature Accuracy (*10)	73.7 ± 3.26
Novelty Score (*11)	0.8 ± 0.02

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6042735042735041	0.6760683760683761	-0.010256410256410253	
F1	0.7971299319768493	0.7208949683315454	0.7924840539605726	-0.00464587801627675	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6974358974358974	0.5555555555555556	0.6871794871794872	-0.010256410256410253	
F1	0.8009226972186992	0.668469273571382	0.799191750913388	-0.0017309463053111873	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6786324786324787	0.7068376068376068	0	
F1	0.8279091624529537	0.796295542911873	0.8279091624529537	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376069	0.6188034188034187	0.7213675213675214	0.014529914529914544	
F1	0.8101504593659687	0.7253460498568405	0.8221400688758574	0.011989609509889999	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.623076923076923	0.7111111111111111	0.00854700854700855	
F1	0.7997903571549948	0.7216190815395159	0.8045918892049443	0.004801532049949464	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5692307692307692	0.7102564102564102	0.006837606837606846	
F1	0.8108734553482932	0.6744104646388743	0.8154002605060395	0.004526805157746128	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5880341880341879	0.6991452991452992	0.024786324786324775	
F1	0.7821343746683586	0.6861369490571143	0.805739733184765	0.023605358516406338	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6581196581196582	0.5709401709401709	0.6598290598290599	0.0017094017094017254	
F1	0.7617684147027333	0.6740876517909291	0.7608638660219154	-0.0009045486808177628	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743591	0.5641025641025641	0.5658119658119658	0.006837606837606846	
F1	0.5587432119380917	0.6654640722896268	0.5725482919316979	0.013805079993606206	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.5683760683760684	0.7008547008547008	0.018803418803418813	
F1	0.7884286132758873	0.6759928138666419	0.799541721989697	0.011113108713809650	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	62	Male	0.6	0.1	160	42	110	4.9	2.6	1.1
2	57	Male	1.0	0.3	187	19	23	5.2	2.9	1.2
3	48	Female	1.1	0.7	527	178	250	8.0	4.2	1.1
4	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
5	24	Male	3.3	1.6	174	11	33	7.6	3.9	1.0
6	41	Male	1.2	0.5	246	34	42	6.9	3.4	0.97
7	34	Female	1.2	0.4	210	30	40	6.8	3.5	1.0
8	29	Male	0.9	0.2	150	20	25	7.0	3.2	1.1
9	45	Female	3.5	1.2	300	50	60	7.5	3.0	0.9
10	52	Male	2.1	0.8	220	40	55	6.2	2.8	1.2
11	60	Female	1.8	0.7	180	35	45	5.9	3.1	1.1
12	41	Male	0.7	0.3	190	25	30	6.0	2.9	1.0
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.5	0.1	162	155	108	8.1	4.0	0.9
2	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
3	19	Female	0.7	0.2	186	166	397	5.5	3.0	1.2
4	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
5	22	Male	0.9	0.3	179	18	21	6.7	3.7	1.2
6	39	Male	0.6	0.2	188	28	43	8.1	3.3	0.6
7	29	Female	0.9	0.3	210	40	30	6.5	3.2	1.1
8	55	Male	2.5	1.0	300	75	50	7.2	3.8	0.9
9	41	Female	1.2	0.5	180	60	45	6.8	3.4	1.0
10	34	Male	0.8	0.2	150	20	25	5.5	2.8	1.2
11	62	Female	3.0	1.5	400	90	70	7.0	3.6	0.8
12	23	Male	1.0	0.4	220	30	20	5.9	3.1	1.3
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Male	0.8	0.2	290	139	87	7.0	3.0	0.7
2	68	Male	0.7	0.1	145	20	22	5.8	2.9	1.0
3	40	Male	1.1	0.3	230	1630	960	4.9	2.8	1.3
4	65	Female	0.7	0.2	182	23	28	6.8	2.9	0.7
5	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
6	26	Male	42.8	19.7	390	75	138	7.5	2.6	0.5
7	29	Female	0.5	0.1	150	22	30	6.5	3.2	0.8
8	45	Male	1.2	0.4	250	40	60	7.0	3.5	1.1
9	34	Female	0.9	0.3	180	25	35	6.8	3.4	0.9
10	50	Male	2.5	1.0	300	50	70	7.5	4.0	1.2
11	60	Female	1.0	0.2	200	30	40	6.0	2.8	0.7
12	22	Male	0.6	0.1	160	20	25	5.5	2.5	0.9
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.7	0.2	265	30	28	5.2	1.8	0.52
2	30	Male	1.6	0.4	332	84	139	5.6	2.7	0.9
3	28	Male	0.6	0.1	177	36	29	6.9	4.1	1.4
4	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
5	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
6	47	Female	0.8	0.2	236	10	13	6.7	2.9	0.76
7	29	Female	0.9	0.3	150	35	22	6.5	3.0	1.2
8	52	Male	1.2	0.4	200	45	30	7.0	3.5	1.0
9	38	Female	0.8	0.2	180	40	25	6.8	3.2	1.1
10	45	Male	1.5	0.5	250	50	35	7.2	3.8	1.3
11	60	Female	1.0	0.3	220	30	20	5.8	2.9	0.9
12	34	Male	0.7	0.2	160	38	28	6.2	3.0	1.0
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Male	32.6	14.1	219	95	235	5.8	3.1	1.1
2	68	Male	1.8	0.5	151	18	22	6.5	4.0	1.6
3	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
4	43	Female	0.9	0.3	140	12	29	7.4	3.5	1.8
5	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
6	60	Male	0.7	0.2	171	31	26	7.0	3.5	1.0
7	29	Female	1.5	0.4	120	22	25	6.8	3.4	1.0
8	34	Male	2.3	0.9	200	30	40	7.5	4.0	1.1
9	45	Female	0.6	0.2	150	18	20	6.0	3.1	0.8
10	52	Male	3.1	1.2	250	35	45	7.2	3.8	1.2
11	39	Female	1.8	0.5	180	25	30	6.5	3.0	0.9
12	60	Male	0.9	0.3	220	40	50	7.0	4.2	1.1
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	41	Female	0.9	0.2	201	31	24	7.6	3.8	1.0
2	46	Male	9.4	5.2	268	21	63	6.4	2.8	0.8
3	62	Male	1.8	0.9	224	69	155	8.6	4.0	0.8
4	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
5	64	Male	1.1	0.5	145	20	24	5.5	3.2	1.39
6	45	Male	1.3	0.6	166	49	42	5.6	2.5	0.8
7	34	Female	1.2	0.4	220	30	28	7.1	3.5	1.02
8	29	Male	0.8	0.2	180	25	20	6.5	3.1	0.93
9	52	Female	2.5	1.1	250	40	35	7.8	4.0	1.05
10	45	Male	1.0	0.5	190	22	30	6.3	2.9	0.85
11	60	Female	3.0	1.5	300	50	45	8.0	3.8	0.95
12	38	Male	1.5	0.7	210	28	25	6.7	3.6	1.1
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06
2	60	Male	5.8	2.7	599	43	66	5.4	1.8	0.5
3	48	Male	1.6	1.0	588	74	113	7.3	2.4	0.4
4	50	Male	1.6	0.8	218	18	20	5.9	2.9	0.96
5	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
6	32	Male	15.0	8.2	289	58	80	5.3	2.2	0.7
7	34	Male	1.2	0.5	210	40	30	6.5	3.0	0.8
8	29	Female	0.9	0.3	150	25	20	5.8	3.2	1.1
9	42	Male	2.5	1.1	300	50	40	7.0	3.5	0.9
10	55	Female	3.1	1.4	220	60	45	6.8	3.1	0.7
11	38	Male	0.8	0.2	180	30	25	5.5	2.8	1.0
12	45	Female	1.5	0.7	250	55	35	6.0	3.4	0.8
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Female	0.7	0.1	240	12	15	7.0	3.0	0.7
2	32	Male	12.7	6.2	194	2000	2946	5.7	3.3	1.3
3	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
4	38	Male	3.7	2.2	216	179	232	7.8	4.5	1.3
5	48	Female	1.0	1.4	144	18	14	8.3	4.2	1.0
6	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
7	29	Female	1.2	0.4	150	25	30	6.5	3.2	0.9
8	34	Male	0.8	0.2	200	30	35	7.0	3.5	1.0
9	45	Female	2.5	1.0	180	40	50	6.8	3.4	1.2
10	52	Male	1.1	0.5	220	35	45	7.5	3.0	0.8
11	60	Female	3.0	1.5	190	28	38	6.0	3.1	0.7
12	25	Male	0.9	0.3	160	22	25	7.2	4.0	1.1
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Female	27.7	10.8	380	39	348	7.1	2.3	0.4
2	45	Male	0.8	0.2	140	24	20	6.3	3.2	1.0
3	18	Male	0.8	0.2	282	72	140	5.5	2.5	0.8
4	30	Male	1.6	0.4	332	84	139	5.6	2.7	0.9
5	38	Male	0.9	0.3	310	15	25	5.5	2.7	1.0
6	21	Male	3.9	1.8	150	36	27	6.8	3.9	1.34
7	29	Female	1.2	0.5	150	30	40	6.0	3.0	1.0
8	54	Male	0.9	0.3	200	45	50	7.0	3.5	1.1
9	39	Female	2.5	1.0	250	60	70	6.5	2.8	0.8
10	45	Male	1.0	0.4	180	35	30	5.5	2.6	0.9
11	62	Female	0.7	0.2	220	50	60	6.8	3.2	1.0
12	34	Male	3.0	1.5	300	70	80	7.2	3.6	1.2
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	1.7	0.8	331	36	53	7.3	3.4	0.9
2	60	Male	0.6	0.1	186	20	21	6.2	3.3	1.1
3	60	Male	6.8	3.2	308	404	794	6.8	3.0	0.7
4	75	Male	2.9	1.3	218	33	37	3.0	1.5	1.0
5	19	Male	1.4	0.8	178	13	26	8.0	4.6	1.3
6	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51
7	34	Female	1.2	0.5	200	45	30	7.0	4.0	1.2
8	29	Male	0.8	0.3	150	25	20	6.5	3.5	1.0
9	52	Male	3.5	1.5	300	60	50	7.5	3.8	0.9
10	41	Female	1.0	0.4	220	35	28	6.8	4.2	1.1
11	38	Male	2.1	0.9	180	40	32	7.2	3.6	1.0
12	45	Female	0.6	0.2	250	30	25	7.1	4.1	1.3
13	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
14	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
15	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
16	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0